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CS 414 Programming Language
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Assignment 03

Step 1: Write a grammar in EBNF format that defines the parsing rules for this command. Identify the tokens, variables, and start symbol for this grammar. Define examples for each of the five supported commands and draw an AST that would be generated from your grammar.

Part 1: Grammar:

```
command = ls_command | cd_command | cat_command | print_command | exec_command;

ls_command = "ls", [folder];

cd_command = "cd", [path];

cat_command = "cat", filename;

print_command = "print", filename;

exec_command = "exec", filename;

filename = name, ".", extension;

name = letter, letter, letter, letter, letter, letter, letter, letter;

extension = letter, letter, letter;

letter = upper_case | lower_case;

upper_case = "A" | "B" | "C" | "D" | "E" | "F" | "G" | "H" | "I" | "J" | "K" | "L"
            | "M" | "N" | "O" | "P" | "Q" | "R" | "S" | "T" | "U" | "V" | "W" | "X"
            | "Y" | "Z";

lower_case = "a" | "b" | "c" | "d" | "e" | "f" | "g" | "h" | "i" | "j" | "k" | "l" | "m" | "n" | "o" |
            "p" | "q" | "r" | "s" | "t" | "u" | "v" | "w" | "x" | "y" | "z";

folder = folder_char, [folder_char], [folder_char], [folder_char], [folder_char],
        [folder_char], [folder_char], [folder_char];

folder_char = letter | digit;

digit = "0" | "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9";

path = folder, {separator, folder};

separator = "\\";
```

Part 2: Tokens, Variables, and Start Symbol

$G = \{V, T, S, P\}$

$V = \{\text{command, ls_command, cd_command, cat_command, print_command, exec_command, filename, name, extension, letter, upper_case, lower_case, folder, folder_char, digit, path, separator}\}$

$T = \{\text{ls, cd, cat, print, exec, A-Z, a-z, 0-9, ., \backslash}\}$

$S = \text{command}$

$P = \{\text{EBNF production rules defined in Part 1}\}$

Part 3: Examples for Each of the Five Supported Commands and AST

1. ls docs
2. cd docs\cs414
3. cat TESTFILE.txt
4. print ReportAA.txt
5. exec RUNAFILE.EXE

ASTs:

